DSP Report (4)

Experiment (4): Digital filters

Design a BPF with the following specifications:

- Order = 10
- Cut-off frequencies = 10000 Hz – 30000 Hz

Where the input signal is:

$$x(t) = 2 * \cos(10000\pi t) + 3 * \cos(30000\pi t) + 5 * \cos(40000\pi t)$$

Plot input and output signals in frequency domain then plot the filter frequency response.